Esercizio 1. (Martina Apicella & Andrea Morello) Let $\varphi(x;z) \in L(A)$. Let $M \supseteq A$ be a sufficiently saturated model. Let say that $p(x) \in S_\varphi(M)$ be invariant over A if it2′. holds for every $a, b \in M^z$. Prove that there is the unique global φ-type that extends $p(x)$ and is invariant over A. Prove that such a type does not depend on the choice of M. What is the required degree of saturation?

Esercizio 2. (Beatrice Calderaro) Let $N \succeq M$ be saturated and of cardinality $> |M|$. Let $a_i \mathop{\downarrow}_M N, b$, for $i = 1, 2$. Prove that if $a_1 \equiv_N a_2$ then $a_1 \equiv_{N,b} a_2$.

Esercizio 3. (Mattia Lo Mele & Daniele Lanciotti) Show that in general there is no type $p(z) \subseteq L(M, a)$ such that

$$a \mathop{\downarrow}_M b \;\Leftrightarrow\; b \models p(z).$$

Hint: consider the theory of dense linear orders without endpoints.

Esercizio 4. (Francesco Dam) Prove that the following are equivalent

1. T has weak elimination of imaginaries
2. $\mathrm{dcl}^{\mathrm{eq}} A \cap \mathrm{dcl}^{\mathrm{eq}} B = \mathrm{dcl}^{\mathrm{eq}}(A \cap B)$ for every algebraically closed sets $A, B \subseteq \mathfrak{U}$.

Esercizio 5. (Filippo Giugie) Let $a \mathop{\downarrow}_M b$. Prove that for every c there is $b' \equiv_{M,a} b$ such that $a, c \mathop{\downarrow}_M b'$.